

# Orbital Rotation Gradient and Hessian for Exponential Parameterization (RDMFT-style, expressed via the Generalized Fock Matrix)

## Conventions

One-particle and two-particle reduced density matrices:

$$D_{ab} = \langle 0 | a_a^\dagger a_b | 0 \rangle, \quad (1)$$

$$\Gamma_{pqrs} = \frac{1}{2} \langle 0 | a_p^\dagger a_q^\dagger a_s a_r | 0 \rangle. \quad (2)$$

Two-electron integrals in physicist's notation,  $\langle ik|jl \rangle$ .

Hamiltonian:

$$\hat{H} = \hat{H}_1 + \hat{H}_2, \quad \hat{H}_1 = \sum_{pq} h_{pq} a_p^\dagger a_q, \quad \hat{H}_2 = \frac{1}{2} \sum_{uvwx} \langle uv|wx \rangle a_u^\dagger a_v^\dagger a_x a_w. \quad (3)$$

Orbital rotation generator (anti-Hermitian):  $\hat{\kappa} = \sum_{pq} \kappa_{pq} a_p^\dagger a_q$ , with  $\kappa_{qp} = -\kappa_{pq}^*$ .

## Generalized Fock matrix

$$F_{qp} = \sum_r D_{rq} h_{rp} + 2 \sum_{rst} \Gamma_{srtq} \langle sr|tp \rangle. \quad (4)$$

## Gradient

$$g_{pq} = F_{qp} - F_{pq}^*. \quad (5)$$

## Hessian

Energy expansion:

$$E(\boldsymbol{\kappa}) \approx E(\mathbf{0}) + \sum_{pq} \kappa_{pq} g_{pq} + \frac{1}{2} \sum_{pq,rs} \kappa_{pq} \kappa_{rs} G_{pq,rs} + \dots, \quad G_{pq,rs} = \langle 0 | [[\hat{H}, a_p^\dagger a_q], a_r^\dagger a_s] | 0 \rangle. \quad (6)$$

## One-electron contribution

$$G_{pq,rs}^{(1)} = \delta_{qr} \sum_u h_{up} D_{us} - h_{sp} D_{rq} - h_{qr} D_{ps} + \delta_{sp} \sum_v h_{qv} D_{rv}. \quad (7)$$

## Two-electron contribution

$$\begin{aligned}
G_{pq,rs}^{(2)} = & + 2 \delta_{rq} \sum_{uvt} \langle uv|pt \rangle \Gamma_{uvst} - 2 \sum_{uv} \langle uv|pr \rangle \Gamma_{uvsq} \\
& + 2 \sum_{ut} \langle us|pt \rangle \Gamma_{ruqt} - 2 \sum_{ut} \langle su|pt \rangle \Gamma_{ruqt} \\
& + 2 \sum_{ut} \langle uq|rt \rangle \Gamma_{pust} - 2 \sum_{uw} \langle uq|wr \rangle \Gamma_{pusw} \\
& + 2 \sum_{wt} \langle sq|wt \rangle \Gamma_{rpwt} - 2 \delta_{sp} \sum_{uwt} \langle uq|wt \rangle \Gamma_{ruwt}.
\end{aligned} \tag{8}$$

## Full Hessian, Fock-matrix form

The  $\delta_{qr}$  and  $\delta_{sp}$  terms of  $G^{(1)}$  and  $G^{(2)}$  combine exactly into  $F_{sp}$  and  $F_{rq}^*$ , leaving six two-electron “exchange-type” terms that do not reduce to  $F$ :

$$\begin{aligned}
G_{pq,rs} = & \delta_{qr} F_{sp} + \delta_{sp} F_{rq}^* - h_{sp} D_{rq} - h_{qr} D_{ps} \\
& - 2 \sum_{uv} \langle uv|pr \rangle \Gamma_{uvsq} + 2 \sum_{ut} \langle us|pt \rangle \Gamma_{ruqt} - 2 \sum_{ut} \langle su|pt \rangle \Gamma_{ruqt} \\
& + 2 \sum_{ut} \langle uq|rt \rangle \Gamma_{pust} - 2 \sum_{uw} \langle uq|wr \rangle \Gamma_{pusw} + 2 \sum_{wt} \langle sq|wt \rangle \Gamma_{rpwt}
\end{aligned} \tag{9}$$

**Status note.** The one-electron block  $G_{pq,rs}^{(1)}$  and the gradient  $g_{pq}$  were cross-checked by more than one derivation route. The two-electron block  $G_{pq,rs}^{(2)}$  (and hence the boxed full Hessian) was obtained by direct operator-algebra expansion but has *not yet* been independently verified numerically (e.g. by a finite-difference check of  $E(\boldsymbol{\kappa})$  against this Hessian for a small test system). Treat it as a first-pass result pending that check before using it in production code.